Yu (Jennie) Zheng

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EDUCATION & ACADEMIC TRAINING

University of California, Berkeley (UC Berkeley) *Feb 2026 - Present*
Associate Specialist & Privacy Engineer, Center for Effective Global Action. Berkeley, US
Topic: Privacy Enhancing Technologies, Large-scale Data Engineering, LLM&Agents, Global Playbook.
Manager/Advisor: Nitin Kohli; Faculty Advisors: Edward Miguel, Joshua Blumenstock.

University of California, Irvine (UC Irvine) *Sep 2024 - Feb 2026*
Postdoctoral Scholar, Department of Electrical Engineering & Computer Science. Irvine, US
Topic: Privacy-Preserving AI, LLM Privacy & Security. Results: [M2],[C8],[C7],[N1], NSF Proposals.
Advisors: Zhou Li (Primary, UCI), Yupeng Zhang (Co-Advisor, UIUC).

Chinese University of Hong Kong (CUHK) *Aug 2018 - Jul 2024*
Doctor of Philosophy, Department of Information Engineering. Hong Kong SAR
Thesis: Communication-Efficient Protocols for Secure Training. Results: [C3],[M1], Productization.
Committee: Kehuan Zhang (Chair), Hongkai Chen, Haibo Hu (PolyU), Sze Ming Chow, Sze Yiu Chau.

Northeastern University (NEU) *Oct 2014 - Jun 2018*
Bachelor of Engineering, Department of Communication Engineering. Shenyang, CN
Thesis: Search over Encrypted Videos. Results: [C1],[J1]. Ranking: 1st of 96.

SELECTED AWARD

Summer Research Institute Fellowship of EPFL, Switzerland (1 of 20).	2023
ACM MMSys Best Student Paper Award (1st Authorship).	2022
Bachelor Dissertation Award (Top 1%).	2018
ICCT Best Paper Award.	2017
National Scholarship of China ×2 (Top 0.2%).	2016/2015

SELECTED PUBLICATIONS & PREPRINTS

Citations: 835; h-index: 12; i10-index: 12; As of May 2026. [Full List]

arXiv Manuscripts († for co-first authorship, # for alphabetical order)

[M2] Danyu Sun, Jinghuai Zhang, Jiacen Xu, **Yu Zheng**, Yuan Tian, and Zhou Li. "From Alerts to Intelligence: A Novel LLM-Aided Framework for Host-based Intrusion Detection." [arXiv]

[M1] **Yu Zheng**†, Qizhi Zhang†, Chenang Li, Lichun Li, Kai Zhou, and Shan Yin. "Secure Graph Convolutional Network on Vertically Split Data from Sparse Matrix Decomposition." Preliminary version accepted in Deep Learning Security and Privacy Workshop co-located with the IEEE Symposium on Security and Privacy, 2025. [Poster] [arXiv] (Industrial Intern & Productizaiton)

Conference

[C9] Zhou Li, **Yu Zheng**, Tianhao Wang, and Sang-Woo Jun. "Network-Aware Differential Privacy." Annual IEEE/IFIP International Conference on Dependable Systems and Networks (DSN), 2026.

[C8] **Yu Zheng**, Chenang Li, Zhou Li, and Qingsong Wang. "Convergent Privacy Framework for Multilayer GNNs through Contractive Message Passing." Network and Distributed System Security Symposium (NDSS), 2026. [Code]

[C7] Jiacen Xu, Chenang Li, **Yu Zheng**, and Zhou Li. "Entente: Cross-silo Intrusion Detection on Network Log Graphs with Federated Learning." Network and Distributed System Security Symposium (NDSS), 2026. [Code]

[C6] Xiuhua Wang, Shikang Li, Fengrui Fan, Shuai Wang, Yiwei Li, and **Yu Zheng**. "Improving Byzantine-resilience in Federated Learning via Diverse Aggregation and Adaptive Variance Reduction." International Conference on Information and Communications Security (ICICS), 2025.

[C5] Xiangfu Song, **Yu Zheng**, Jianli Bai, Changyu Dong, Zheli Liu, and Ee-Chien Chang. "DISCO: Dynamic Searchable Encryption with Constant State." ACM Asia Conference on Computer and Communications Security (AsiaCCS), 2024. [Code]

[C4] Zhiqin Yang, Yonggang Zhang, **Yu Zheng**, Xinmei Tian, Hao Peng, Tongliang Liu, and Bo Han. "FedFed: Feature Distillation against Data Heterogeneity in Federated Learning." Advances in Neural Information Processing Systems (NeurIPS), 2023. [Code]

[C3] **Yu Zheng**†, Qizhi Zhang†, Sherman S.M. Chow, Yuxiang Peng, Sijun Tan, Lichun Li, and Shan Yin. "Secure Softmax/Sigmoid for Machine-Learning Computation." Annual Computer Security Applications Conference (ACSAC), 2023. [Code] [Slides] (Industrial Intern & Productizaiton)

[C2] **Yu Zheng**, Wei Song, Minxin Du, Sherman S.M. Chow, Qian Lou, Yongjun Zhao, Xiuhua Wang. "Cryptography-Inspired Federated Learning Variants for GAN and Meta Learning." International Conference on Advanced Data Mining and Applications (ADMA), 2023. [Code] [Slides]

[C1] **Yu Zheng**, Heng Tian, Minxin Du, and Chong Fu. "Encrypted Video Search: Scalable, Modular, and Content-similar." ACM Multimedia Systems Conference (MMSys), 2022. (***Best Student Paper Award***) [Code] [Slides]

Journal

[J2] Xing Ai, Yulin Zhu, **Yu Zheng**, Gaolei Li, Jianhua Li, and Kai Zhou. "Revisiting Adversarial Robustness of GNNs against Structural Attacks: a Simple and Fast Approach." Transactions on Information Forensics & Security (TIFS), 2025.

[J1] **Yu Zheng**, Wenchao Zhang, Wei Song, Xiuhua Wang, and Chong Fu. "Encrypted Video Search with Single/Multiple Writers." ACM Transactions on Multimedia Computing, Communications, and Applications, 2024. (Invited Submission)

Non-arXiv Manuscripts

[N1] Chenang Li, **Yu Zheng**, Tianyi Liu, Yupeng Zhang, and Zhou Li. "EAGLE: Efficient Privacy-Preserving Graph-based Anomaly Detection with Practical Cross-silo Partitioning."

RESEARCH GRANTS & PROPOSALS

- "User-Customized Privacy Frameworks for LLM Usage: Optimizing Utility-Privacy Frontier via Differential Privacy and Secure Computation." from NSF SaTC Aspiring PI Workshop, 2025. Investigator
- "Collaborative Research: PDaSP: Track 1: Practical Secure Multiparty Computations for Graph-based Intrusion Detection Systems." from **NSF**, 2025-2027. (**Grant: $750K**) I contributed core techniques [M1] and drafted the proposal with two advisors. I am mentoring a new PhD student for [N1].
  Co-PIs: Zhou Li (UC Irvine), Yupeng Zhang (UIUC). Senior Personnel
- "Efficient Privacy-Enhancing Techniques for Edge Computing." from NSF NeTS Early-Career Investigators Workshop, 2025. (Travel Grant: ≈$1K) Investigator
- "Secret Communications through Steganography and Chaotic Encryption." from NEU Key Program for Undergraduate Research Training, 2016 - 2017. (Award: ¥86K≈$13K) Team Leader

PROFESSIONAL SERVICE

Organizing Committee:
(*26'*): Publicity Chair for PRISM Workshop co-located with NDSS. San Diego

Program Committee:
(*27'*): S&P.
(*26'*): ACM CCS, S&P (Travel Grant).
(*25'*): ACSAC, AAAI, ACNS-SiMLA, ACM WPES@CCS, ICNC.
(*24'*): CCS AE, NDSS AE (Discussion Lead), TheWebConf AE, USENIX Security AE, EAI ICECI.
(*23'*): CCS AE, AJCAI.

Session Chair/Host:
(*25'*): S&P (San Francisco), RSA (San Francisco), NDSS (San Diego).
(*24'*): AsiaCCS (Singapore).

Review Service:
ICLR, TDSC. (*26'*)
ACM TOPS, ICLR, ICML, JCS, NeurIPS, TMLR, TDSC. (*25'*)
AISTATS, AsiaCCS, ICLR, ICML, IEEE TDSC, Information Sciences, NeurIPS. (*24'*)
Information Sciences, NCAA, NeurIPS, JMANS. (*23'*)

TEACHING SERVICE

Mentoring:
Chenang Li, PhD@UCI (Advisors: Zhou Li@UCI, Yupeng Zhang@UIUC) Oct 2024 - Dec 2025
Topic: System Security, Secure Graph Learning. Result: [N1].

Guest Lectures:
EECS231(by Zhou Li)@UCI: MPC Protocols for Sparse Graph Computations. Feb 2025
CAP6614(by Qian Lou)@UCF: Secure Machine Learning with Multiparty Computation. Feb 2024

Teaching Assistant Courses:
Introduction to Cyber Security. Spring 2020, Spring 2019
Web Programming and Security. Spring 2021, Fall 2019
Electronic Circuit Design Lab. Fall 2021, Fall 2020, Fall 2018
Microcontroller and Embedded Systems Lab. Spring 2022, Fall 2022
Technology Strategy and Commercialization. Spring 2024
Introduction to Microcontroller. Summer 2015

TALKS & PRESENTATION

(Colloquium): Advancing DP and Secure Compt.: FDN. for Trustworthy GNNs. CS@UCR, Jan 2026
(Poster): Secure GCN on Vertically Split Data. GEECS Annual Technology Showcase, Jun 2025
(Poster): Secure Graph Convolutional Network on Vertically Split Data. DLSP@S&P, May 2025
(Seminar): Secure Graph Convolutional Network on Vertically Split Data. ICS@UCI, Feb 2025
(Pitch): Practical Privacy-Enhancing Techniques for Edge Computing. NeTS@NSF, Jan 2025
(Talk): Secure Graph Convolutional Network on Vertically Split Data. ECE@VirginiaTech, Jan 2025
(Seminar): Secure Graph Convolutional Network on Vertically Split Data. CS@UMD, Jan 2025
(Poster): Secure Graph Convolutional Network on Vertically Split Data. WiCyS@UCI, Nov 2024
(Seminar): Efficient MPC Protocols for Secure Machine Learning. CSE@UCSC, Apr 2024
(Conference): Secure Softmax/Sigmoid for Machine-Learning Computation. ACSAC, Dec 2023
(Oral/Conference): Cryptography-Inspired Federated Learning Variants. ADMA/NEU, Aug 2023
(Poster): Secure Softmax/Sigmoid for Machine-Learning Computation. EPFL, Jul 2023